Dheeraj Nallamotu

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Education

▸ **KL University** — Guntur, Andhra Pradesh, India
B.Tech. in Computer Science and Engineering (Honors) — *Aug 2023 – May 2027 (Expected)*
Specialization in Artificial Intelligence
CGPA: 9.34/10
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Machine Learning, Deep Learning, Natural Language Processing

▸ **Narayana Junior College** — Vijayawada, Andhra Pradesh, India
Intermediate, MPC (Mathematics, Physics, Chemistry) — *May 2021 – Apr 2023*
CGPA: 9.68/10

Research and Projects

▸ **FreshGuard: Human-in-the-Loop Decision Support for Shelf-Life Monitoring** — *Team Lead & Developer*
First-author paper accepted at ICACTCE '26 (Paper ID 128)
Python, FastAPI, SQLAlchemy, SQLite, HTML/CSS/JavaScript

- Led the team and developed the working web prototype for environmental risk assessment of perishable-food batches, with a FastAPI backend, SQLAlchemy/SQLite persistence, and an HTML/CSS/JavaScript interface.
- Implemented rule-based environmental-stress assessment over simulated temperature and humidity readings, classifying each batch as Healthy, Warning, or Critical.
- Implemented alert deduplication that reuses a batch's existing unresolved alert of the same severity instead of creating a new one, and a human review step to Accept, Reject, or Escalate each alert.
- Evaluated the prototype on 4 sample batches and 5,961 simulated sensor readings; the evaluation snapshot recorded 8 alert records (1 active Critical, 3 active Warning, 4 resolved).

▸ **Human-in-the-Loop Escalation Architecture for NLU Deployments (B.Tech. Capstone)** — *Team Lead*
Python, FastAPI, Pydantic, SQLAlchemy, PostgreSQL/SQLite, Uvicorn

- Leading the capstone team building a system that evaluates conversational requests and decides whether to PROCEED with automated processing or ESCALATE to a human reviewer, designed to support travel/airline, food-ordering, e-commerce, and appointment-scheduling domains.
- Built the FastAPI backend and decision-making logic, combining confidence checks, operation-risk checks, and critical-entity validation to determine when a request requires human review.
- Implemented the human-review workflow; contributed to SQLAlchemy database integration and to connecting backend APIs with the frontend for request evaluation, escalation handling, and evaluation history.

▸ **Collaborative Agents Using CrewAI** — *Developer*
Python, CrewAI, LangChain Agents

- Built a multi-agent workflow in which a coordinating (master) agent delegates tasks to specialized agents for content generation, question answering, and summarization, and integrates their responses into a single output.

Experience

▸ **TrekVerse** — Remote
Python Full Stack Development Intern — *Jun 2026 – Sep 2026*
Python, FastAPI, Pydantic, SQLAlchemy, SQLite, Jinja2, HTML/CSS/JavaScript

- Built a FastAPI task-management web application with CRUD REST endpoints returning structured JSON responses.
- Persisted data with SQLAlchemy on SQLite; enforced input correctness with Pydantic request validation, exception handling, and appropriate HTTP status codes.
- Developed a Jinja2, HTML, CSS, and JavaScript frontend integrated with the backend APIs.

Technical Skills

- **Programming & Backend:** Python, SQL, JavaScript, FastAPI, REST APIs, Pydantic, SQLAlchemy, Uvicorn
- **AI/ML & NLP:** Generative AI, Large Language Models (LLMs), NLP, Retrieval-Augmented Generation (RAG)
- **AI Agents:** Multi-Agent Systems, CrewAI, LangChain Agents
- **Databases:** PostgreSQL, SQLite, MongoDB, ChromaDB
- **Web Development:** HTML, CSS, JavaScript, Jinja2, Streamlit
- **Cloud & Tools:** AWS, Oracle Cloud Infrastructure, Microsoft Azure, Git, GitHub

Certifications

- AWS Certified Data Engineer – Associate
- AWS Certified Cloud Practitioner
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- Microsoft Certified: Azure Fundamentals (AZ-900)